

Trainer Debrief — Lab 14

Baseline Capability — Cp, Cpk, Pp, Ppk and Non-Normal Data

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Establish baseline capability for your capstone, handling non-normal data correctly (A4, K2).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-14-baseline-capability.xlsx

BurstPressure: mean ~214.0, overall SD ~21.5. $Pp = (260-180)/(6 \times 21.5) = 0.62$; $Ppk = \min(260-214, 214-180)/(3 \times 21.5) = 0.53$. Cpk (within) is slightly higher than Ppk, which tells you the process is not stable over time. Roughly 5.6% of units fall outside spec — about 2.9 sigma. THIS IS THE BASELINE the whole project improves; Lab 27 re-measures it at Ppk ~1.40. CycleTime is lognormal (Anderson-Darling $p < 0.005$): do NOT compute Ppk on the raw data — transform (Box-Cox, $\lambda \sim 0$) or fit a lognormal distribution first.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

You have confirmed stability before computing capability, tested and documented normality with a p-value, handled non-normality by transformation or distribution fitting, and reported both Cpk and Ppk with an explanation of the gap.

Where this sits in the capstone

Re-derive capability analysis in full, then go beyond Green Belt: the distinction between Cpk and Ppk that most practitioners get wrong, and what to do when the data is not normal — which is most of the time in cycle-time and defect data. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27

Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30
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If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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